# Aaron Slipper

Department of Mathematics, Duke University

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## Academic appointment

2024–present **Phillip Griffiths Assistant Research Professor of Mathematics**
Duke University. Research mentor: Jayce Getz. Appointment through August 2027.

## Education

2024 **Ph.D. in Mathematics**, University of Chicago
Advisors: Ngô Bảo Châu and Kazuya Kato.
Thesis: *Non-Abelian Fourier Transforms and Normalized Intertwining Operators over Finite Fields for General Parabolic Subgroups*.

2018 **B.A. in Mathematics**, Harvard University, *magna cum laude*
Thesis advisor: Noam Elkies.
Thesis: *Modular Magic: The Theory of Modular Forms and the Sphere Packing Problem in Dimensions 8 and 24*.

## Research interests

Number theory and geometric representation theory; nonlinear Fourier transforms and the Langlands program; logarithmic geometry.

## Publications and preprints

[1] *Bosonic codes from compact phase spaces*.
With D. Roberts, A. Parhizkar, V. V. Albert, and M. Hafezi. Joint first authorship with D. Roberts.
2026. arXiv:2608.31156.

[2] *Quasi-Classical Braverman–Kazhdan Intertwiners via Quiver Varieties*.
With N. Grantcharov.
2026. arXiv:2606.14637.

[3] *Geometrization of the Schrödinger Model for the Minimal Representation of an Even Orthogonal Group: The de Rham Setting*.
2026. arXiv:2604.11023.

[4] *The ρ-Fourier transform*.
With J. R. Getz, A. Gutiérrez Terradillos, F. Hosseinijafari, G. Xi, H. Yao, and A. Zhao.
2025; revised 2026. arXiv:2512.00182.

[5] *Modulation groups*.
With J. R. Getz, A. Gutiérrez Terradillos, F. Hosseinijafari, B. Hu, S. Lee, M.-H. Tomé, H. Yao, and A. Zhao.
2025. arXiv:2510.23932.

[6] *Logarithmic Geometry and Geometric Class Field Theory*.
2025; revised 2026. arXiv:2508.08648.

[7] *Parabolic Kazhdan–Laumon and the Kloosterman Fourier Transform for Quadric Cones*.
2024; published in *Algebras and Representation Theory* (2026). arXiv:2410.17476.

## Work in progress

- *Quantized Braverman–Kazhdan Reflection Functors*.
  With Nikolay Grantcharov. In preparation.
- *Geometrization of the Schrödinger Model for the Minimal Representation of the Even Orthogonal Group: the Étale Setting*. In preparation.

## Awards and fellowships

2025–2026 **Lewis Blake Award for Excellence in Teaching**
Duke University.

2025 **Recognition for teaching excellence**, Trinity College, Duke University
Spring 2025 MATH 242 evaluations ranked in the top 5% of comparable Natural Sciences undergraduate courses in at least two evaluation categories.

2020 **Physical Sciences Teaching Prize**, University of Chicago

2018 **NSF Graduate Research Fellowship**

2018 **Phi Beta Kappa**, Alpha of Massachusetts

2017, 2018 **Derek Bok Certificate of Distinction in Teaching**, Harvard University

2017 **Harvard College Scholar**

2016 **Wendell Prize Scholarship finalist**, Harvard University

2015 **Detur Book Prize**, Harvard University

## Research mentoring

Fall 2026 **Supervised reading course and graduate seminar**
Supervising Kai Wang in algebraic stacks and $\mathrm{Bun}_G$; the course has developed into a private seminar for 5–6 early graduate students on algebraic geometry, stacks, and the moduli space of principal $G$-bundles.

2025 **Co-advisor, senior honors thesis of Marie-Hélène Tomé**
Duke University, with Jayce Getz.
*Towards a Geometric Understanding of Boundary Terms in the Poisson Summation Formula for Quadric Cones*.
Recipient of Duke's 2025 Excellence in Mathematical Research Award.

## Professional service and research programs

**Journal refereeing:** *Selecta Mathematica*; *International Mathematics Research Notices* (IMRN).

2024–2027 **Organizer**, Duke Number Theory Seminar

2024–2025 **Participant**, Duke Research Scholars
*The Poisson summation conjecture*. Mentor: Jayce Getz.

2023–2024 **Co-founder and co-organizer**, BunG Seminar
University of Chicago. Founded with Nikolay Grantcharov in Winter 2023.

## Teaching experience

### Duke University

*Course instructor, except where noted below.*

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| Spring 2027 | **Commutative Algebra and Algebraic Geometry**, MATH 602<br>Scheduled. |
| Fall 2026 | **Representation Theory**, MATH 603 / PHYSICS 603<br>Current. |
| Spring 2026 | **Introduction to Abstract Algebra**, MATH 401 / MATH 701 |
| Fall 2025 | **Number Theory**, MATH 305S |
| Spring 2025 | **Discrete Mathematics and Proofs**, MATH 242 / COMPSCI 232 |
| Fall 2024 | **Multivariable Calculus for Economics**, MATH 202D<br>Discussion leader. |

### University of Chicago

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| Fall 2021, 2023 | **Calculus II**, MATH 15200. Instructor. |
| Spring 2023 | **Paris Mathematics Study Abroad Program**<br>Teaching assistant: Algebraic Curves (Ngô Bảo Châu), $p$-adic Harmonic Analysis (John Boller), Introduction to Dynamical Systems (Simion Filip). |
| Spring 2022 | **Linear Algebra**. Instructor. |
| Winter 2022 | **Calculus III**, MATH 15300. Instructor. |
| 2019–2020 | **Inquiry-Based Learning Calculus**<br>College Fellow; course assistant for Sarah Ziesler. |

### Harvard University

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| 2015–2017 | **Discrete Mathematics**, MATH 152. Upper-level course assistant. |

## Selected conferences and schools attended

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| July 14–<br>Aug. 1, 2025 | Summer Research Institute in Algebraic Geometry.<br>Colorado State University, Fort Collins, Colorado. |
| July 22–26, 2024 | WARTHOG: Coherent–Constructible Equivalences in Local Geometric Langlands and Representation Theory. University of Oregon. |
| July 1–19, 2024 | Second Simons Math Summer Workshop: Moduli.<br>Simons Center for Geometry and Physics, Stony Brook University. |
| June 10–14, 2024 | Los Angeles Workshop on Representations and Geometry (LAWRGe):<br>Relative Langlands Duality. University of Southern California. |
| June 2024 | Summer School and Workshop on Relative Langlands Duality.<br>University of Minnesota. |
| Mar. 2–6, 2024 | Arizona Winter School: Abelian Varieties. University of Arizona. |
| Mar. 7–11, 2020 | Arizona Winter School: Nonabelian Chabauty. University of Arizona. |

## Selected talks

### Research seminars and conference talks

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| 2026 | Number Theory Seminar, Duke University<br>Scheduled. |
| Oct. 11, 2026 | *Geometrization of the Minimal Representation of the Even Orthogonal Group*<br>AMS Fall Southeastern Sectional Meeting, Kennesaw State University. |
| Oct. 4, 2026 | *Geometrization of the Minimal Representation of the Even Orthogonal Group*<br>AMS Fall Eastern Sectional Meeting, George Washington University. |
| Sept. 17, 2026 | *Minimal representations, nonlinear Fourier transforms, and boundary geometry*<br>Automorphic Forms and Representation Theory Seminar, Purdue University. |
| Aug. 21, 2026 | *Geometrization of the Minimal Representation of the Even Orthogonal Group*<br>Geometric Methods in Representation Theory Seminar, University of North Carolina at Chapel Hill. |
| Oct. 9, 2024 | *Normalized Intertwining Operators for General Parabolics over a Finite Field*<br>Number Theory Seminar, Duke University. |
| Feb. 10, 2024 | *Normalized Intertwining Operators for General Parabolics over a Finite Field*<br>Junior Number Theory Days, Johns Hopkins University. |
| May 9, 2022 | *An Overview of Geometric Class Field Theory*<br>Junior Number Theory Seminar, University of Oxford. |

### Selected expository seminar lectures, University of Chicago

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| Mar. 27, 2024 | *Shtukas and V. Lafforgue's Proof of Global Langlands for Function Fields*<br>BunG Seminar. |
| Jan. 3, 2024 | *An Overview of the Tamagawa Number Theorem for Function Fields*<br>BunG Seminar. |
| Oct. 11, 2023 | *The Geometric Satake Correspondence*<br>BunG Seminar. |
| Oct. 6, 2023 | *Normalized Intertwining Operators over Finite Fields*<br>Ngô's Langlands Program Seminar. |
| Oct. 2 and 9, 2023 | *On the Bernstein Center for p-adic Groups*<br>Emerton's Categorical Langlands Seminar. |
| Mar. 1, 2023 | *A Mosaic of Examples and Basic Properties of BunG*<br>BunG Seminar. |
| Jan. 2023 | *Braverman–Kazhdan Normalized Intertwining Operators*<br>Ngô's Langlands Program Seminar. |
| Nov. 2019 | *Deligne–Lusztig Theory*<br>Introductory lecture, Ngô's seminar on character sheaves. |

## Languages and additional activities

Spanish (fluent). Classical music: piano, violin, viola, and composition.

Hasty Pudding Theatricals (2015–2018); Harvard–Radcliffe Gilbert and Sullivan Society.